

Homework 3

1.5.4 Let g be in $L^p(\mathbb{R}^n)$ for some p satisfying $0 < p < 1$. Show that

$$\lim_{\delta \rightarrow 0} \frac{1}{|B(x, \delta)|} \int_{B(x, \delta)} |g(y) - g(x)|^p d\lambda(y) = 0$$

for almost all $x \in \mathbb{R}^n$

Proof. For every rational number a there is a set E_a of Lebesgue measure zero such that for $x \in \mathbb{R}^n \setminus E_a$, we have

$$\lim_{\delta \rightarrow 0} \frac{1}{|B(x, \delta)|} \int_{B(x, \delta)} |g(y) - a|^p d\lambda(y) = |g(x) - a|^p$$

. Because g is in $L^p(\mathbb{R}^n)$, the map $h(y) = |g(y) - a|^p$ is integrable over compact domains, and thus h is in $L^1_{\text{loc}}(\mathbb{R}^n)$. Note that there exist multiple enumerations of the rationals. For this case, the specific enumeration does not matter, only that it exists. We can take the union of all E_a , and use subadditivity of measure and the fact that the rationals are countable to get that:

$$\lambda(E) = \lambda\left(\bigcup_{a \in \mathbb{Q}} E_a\right) \leq \sum_{i \in \mathbb{N}} 0.$$

Therefore, for all $a \in \mathbb{Q}$, $x \in \mathbb{R}^n \setminus E$ the condition above holds. Let $r \in \mathbb{R}$. Note that by the reverse triangle inequality and the concavity of $|g(y) - a|^p$, we have that

$$\begin{aligned} ||g(y) - a|^p - |g(y) - r|^p| &\leq ||g(y) - a| - |g(y) - r||^p \\ &\leq |g(y) - a - g(y) + r|^p \\ &= |r - a|^p. \end{aligned}$$

So,

$$\begin{aligned} 0 &\leq \left| \int_{B(x, \delta)} |g(y) - a|^p - |g(y) - r|^p d\lambda(y) \right| && \leq \int_{B(x, \delta)} |r - a|^p d\lambda. \\ 0 &\leq \left| \int_{B(x, \delta)} |g(y) - a|^p d\lambda(y) - \int_{B(x, \delta)} |g(y) - r|^p d\lambda(y) \right| && \leq \int_{B(x, \delta)} |r - a|^p d\lambda. \\ 0 &\leq \left| \frac{1}{|B(x, \delta)|} \int_{B(x, \delta)} |g(y) - a|^p d\lambda(y) - \frac{1}{|B(x, \delta)|} \int_{B(x, \delta)} |g(y) - r|^p d\lambda(y) \right| && \leq |r - a|^p. \end{aligned}$$

We can now split up the absolute value.

$$\begin{aligned} \frac{1}{|B(x, \delta)|} \int_{B(x, \delta)} |g(y) - a|^p d\lambda(y) - |r - a|^p &\leq \frac{1}{|B(x, \delta)|} \int_{B(x, \delta)} |g(y) - r|^p d\lambda(y) \\ &\leq \frac{1}{|B(x, \delta)|} \int_{B(x, \delta)} |g(y) - a|^p d\lambda(y) + |r - a|^p. \end{aligned}$$

Notice that we can take the limsup as δ goes to 0. Because we have shown that $\lim_{\delta \rightarrow 0} \frac{1}{|B(x, \delta)|} \int_{B(x, \delta)} |g(y) - a|^p d\lambda(y) = |g(x) - a|^p$, we can replace the limsup with this value, which gives:

$$-|r - a|^p + |g(x) - a|^p \leq \limsup_{\delta \rightarrow 0} \frac{1}{|B(x, \delta)|} \int_{B(x, \delta)} |g(y) - r|^p d\lambda(y) \leq |r - a|^p + |g(x) - a|^p.$$

If we let $r = g(x)$ and $a \rightarrow g(x)$ by density of $\mathbb{Q}$ in $\mathbb{R}$, we have that

$$0 \leq \limsup_{\delta \rightarrow 0} \frac{1}{|B(x, \delta)|} \int_{B(x, \delta)} |g(y) - g(x)|^p d\lambda \leq 0.$$

Because the limit is nonnegative and bounded above by the limsup, we have shown that

$$\lim_{\delta \rightarrow 0} \frac{1}{|B(x, \delta)|} \int_{B(x, \delta)} |g(y) - g(x)|^p d\lambda = 0.$$

□